सेंट्रल ट्रांसमिशन यूटिलिटी ऑफ इंडिया लिमिटेड
(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)
(भारत सरकार का उद्यम)
CENTRAL TRANSMISSION UTILITY OF INDIA LTD.
(A wholly owned subsidiary of Power Grid Corporation of India Limited)
(A Government of India Enterprise)

Ref: CTU/N/00/CMETS_NR/20

Date: 12-07-2023

As per distribution list

Subject: 20th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

Dear Sir/Ma'am,

Please find enclosed the minutes of the 20th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 30th June, 2023 (Friday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,

12/07/23

(Kashish Bhambhani)
General Manager (CTU)

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Minutes of 20th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 30.06.2023

The 20th Consultation Meeting for Evolving Transmission Schemes in Northern Region (CMETS-NR) was held through VC on 30/06/2023. List of participants is enclosed at **Annexure-X**.

A. Confirmation on the minutes of the 19th CMETS-NR meeting

It was informed that the Minutes of the 19th CMETS-NR meeting held on 31/05/2023 were issued vide letter dated 09/06/2023.

No comments were received from any applicant, accordingly, minutes were confirmed as circulated. The detailed agenda was discussed as below:

B. Application related matters in Northern Region

- It was stated that Hon'ble CERC vide notification dated 01-04-2023 has issued First Amendment of GNA Regulations and notified that the effective date of Regulation 37.1 to 37.8 shall be 05-04-2023 in place of 15-10-2022.
- After scrutiny of the GNA transition options (opted/not opted) received by CTUIL, it has been observed that some of the applicants have surrendered the connectivity and /or LTA at Bhadla-II, Fatehgarh-II, Fatehgarh-III PS in NR. Further, Connectivity granted at Bikaner-II PS of 675 MW was revoked by CTUIL vide letter dated 05.04.2023. The details regarding margins released is already updated in CTUIL website. The bays vacated at these substations post surrender/revocation were proposed to be allotted to other RE developers granted connectivity/LTA in a new pooling station in proximity to these Pooling stations based on application priority.
- It was informed that in this regard, a meeting was held on 20.06.2023 with SECI, RE developers granted connectivity/LTA at Bhadla-III, Fatehgarh-IV & Bikaner-III PS, wherein the reallocation of bays at Bhadla-II, Bikaner-II & Fatehgarh-III PS was deliberated (MoM awaited). In the meeting, SECI also highlighted GIB related issues for applicants at Fatehgarh-II PS and their willingness to relocate to proximal station on vacated bays.
- During the present meeting, SECI reiterated that the generation projects who have been granted connectivity at Fatehgarh-II S/s and awaiting from the GIB committee would also be included for reallocation for vacated margin at Fatehgarh-III PS. Regarding this, CTUIL stated that GIB committee decision has not been received yet for the developers at Fatehgarh-II PS & CTUIL cannot change the connectivity granted to these applicants on Suo-moto basis as their bays are already under advanced stage of implementation. CTUIL also informed that in view of this, only provisional reallocation to Fatehgarh-IV PS applicants was carried out in the reallocation

meeting on 20.06.23 and Connectivity grantees at Fatehgarh-II PS were not considered for reallocation. SECI informed that they have highlighted this issue in a recent meeting taken by Secretary (MNRE), wherein, it was informed that suitable directions shall be issued in this regard (minutes awaited).

- Further, it was deliberated that as the above reallocation shall have impact on the Transmission system for GNA for RE applicants, upon completion of reallocation, transition and new cases shall be taken up in July'23 meeting.

1. Applications for Connectivity to ISTS under CERC (Connectivity and General Network Access to the inter-State Transmission System) Regulations, 2022:

It was noted that the following 9 Nos. of applications for Connectivity to ISTS were received in the month of May'23:

TABLE 1

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought (MW)/date	Nature of Applicant	Generation Schedule	Criterion for applying	Connectivity location (requested)
1.	2200000054	Green Infra Clean Wind Technology Limited	Barmer distt., Rajasthan	15.05.2023	300/ 01.06.2025	Generator (Solar)	01.06.2025	Land Route	Fatehgarh-IV PS
2.	2200000067	Utkrrisht Solar Energy Private Limited	Barmer distt., Rajasthan	26.05.2023	300/ 01.04.2026	Renewable Power Park developer (Solar)	01.04.2026	Land Route	Fatehgarh-IV PS
3.	2200000073 (Enhancement)	Juniper Green Cosmic Private Limited	Bikaner distt., Rajasthan	24.05.2023	150/ 31.01.2025	Generator (Solar)	50 MW- 31.01.2025 100 MW- 31.03.2025	Land Route	Bikaner-II PS

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought (MW)/date	Nature of Applicant	Generation Schedule	Criterion for applying	Connectivity location (requested)
4.	2200000034 (Enhancement)	Sprng Pavana Urja Private Limited	Barmer distt., Rajasthan	04.05.2023	150/ 31.12.2026	Generator (Solar)	50 MW- 31.12.2026 100 MW- 30.06.2027	Land Route	Fatehgarh-IV PS
5.	2200000037	Project Nine Renewable Power Private Limited	Bikaner distt., Rajasthan	05.05.2023	450/ 11.06.2025	Generator (Solar)	50 MW- 01.03.2025 50 MW- 31.03.2025 75 MW- 15.04.2025 75 MW- 30.04.2025 75 MW- 15.05.2025 75 MW- 31.05.2025 50 MW- 11.06.2025	Land BG Route	Bhadla-III PS
6.	2200000062	Juniper Green India Six Private Limited	Jodhpur distt., Rajasthan	22.05.2023	300/ 31.12.2027	Renewable Power Park developer (Solar)	150 MW- 31.12.2027 150 MW- 30.06.2028	Land BG Route	Bhadla-III PS

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought (MW)/date	Nature of Applicant	Generation Schedule	Criterion for applying	Connectivity location (requested)
7.	2200000063	Juniper Green Stellar Private Limited	Barmer distt., Rajasthan	22.05.2023	300/ 31.12.2026	Renewable Power Park developer (Solar)	150 MW- 31.12.2026 150 MW- 30.06.2027	Land BG Route	Fatehgarh-IV PS
8.	2200000060	Sprng Urja Private Limited	Bikaner distt., Rajasthan	10.05.2023	200/ 31.12.2027	Generator (Solar)	31.12.2027	Land BG Route	Bhadla-III PS
9.	2200000065	Sprng Akshaya Urja Private Limited	Barmer distt., Rajasthan	18.05.2023	100/ 30.06.2025	Generator (Solar)	30.06.2025	LOA or PPA	Fatehgarh-IV PS

For Connectivity applications indicated at **Table-1**, it was deliberated that the optimal transmission system requirement for above applications could only be assessed after completion of GNA transition of applicants granted connectivity under 2009 Regulations as well as reallocation process. In view of this, the above applications shall be discussed in the CMETS/Separate meeting with detailed transmission system required for grant of Connectivity in line with GNA Regulations subsequently.

2. Applications for GNA/GNA_{RE} under CERC (Connectivity and General Network Access to the inter-State Transmission System) Regulations, 2022:

Sl. No.	Application ID	Name of the Applicant	Submission Date	Region	Nature of applicant	GNA within Region (MW)	GNA outside Region (MW)	Total Quantum (MW) of GNA Required	Start Date of GNA	End date of GNA
1.	2200000059	Hindustan Zinc Limited	22-05-2023	NR	Bulk consumer seeking to connect to ISTS	250.0	250.0	500.0 Revised: (250 MW)	31-03-2025	30-03-2050

It was deliberated that M/s HZL has applied for drawl of 500 MW under GNA (250 MW-within region & 250 MW-outside region) at Chittorgarh 400/220 kV ISTS station in Northern region for their Chanderiya Lead Zinc smelter plant from Chittorgarh S/s.

CTUIL enquired about the details regarding total power to be drawn. On this, M/s HZL informed that they intend to draw power from 250 MW RTC project. Solar part of 250 MW shall come from within the region, i.e., Rajasthan. Wind part (250 MW) shall come from outside the region, i.e. Maharashtra. On overall basis, they require only 250 MW power. However, in the NSW application the pre-defined formula for total GNA has added the power from within & outside region (250MW+250MW) and they require GNA for total 250 MW of power only. They had also informed the same vide their clarification letter dated 08/05/23 that they shall draw total 250 MW from either Rajasthan (Within Region) or Maharashtra (Outside Region).

CTUIL informed that at one point of time, the total power intended to be drawn is 250 MW from HZL (Chanderia plant in NR) and for this, constraints may arise outside the region and not inside the region and the total transmission system need to be planned for 250 MW only not 500 MW. However, application is for 500 MW, which is in contradiction to intended requirement of 250 MW. Regarding this, CTUIL also enquired from GRID INDIA about the how scheduling of this 250 MW shall take place. GRID INDIA informed that power in GNA shall be scheduled in line with the PPA for Solar & Wind power and as per the available margin in the Grid at that time. However, for the capacity exceeding granted GNA quantum, T-GNA may be applied by the applicant, if required.

Further, M/s HZL submitted they have proposed to be connected through 400 kV S/c line at Chittorgarh substation and further, they shall be connected at 132 kV level at HZL Switchyard at Chanderiya. For the same M/s HZL shall also construct a pooling station at 400/132 kV level. Regarding this, CTUIL highlighted that proposed 400 kV S/c line for 250 MW shall sub optimal utilizate a 400 kV transmission asset, further with 400kV s/c line there will not be any N-1 reliability for a bulk Consumer. Therefore, it was proposed that M/s HZL may explore feasibility of getting connected at Neemuch substation at 220 kV level, which is also in proximity. HZL agreed to explore the possibility of Connectivity at Neemuch S/s. Space availability for 220kV switchyard at 765/400kV Chittorgarh S/s is also under discussion with POWERGRID.

Further, it was deliberated that while applying for GNA, M/s HZL has proposed the disconnection from STU (RVPNL) after availing GNA under ISTS. For the same, M/s HZL informed that they have submitted the application to STU for grant of NOC for disconnection of 250 MW already granted them. However, the NOC is still awaited from STU.

Accordingly, it was decided that the GNA application from M/s HZL shall be discussed again based on inputs of HZL on above as well as its application treatment.

C. <u>ISTS Expansion</u>

1. Non-Compliance of N-1 contingency in ICTs at Allahabad and Mainpuri Sub stations

It was stated that in 207th OCC meeting held on 19.05.23, POWERGRID agenda (as per their letter dated 20.04.23) for Non-Compliance of N-1 contingency in ICTs at Allahabad and Mainpuri Sub stations was deliberated.

In the meeting, POWERGRID highlighted that at present there are 3x315 MVA ICTs at Allahabad S/s and 2x315 MVA + 1x500 MVA ICTs at Mainpuri S/s and loading on these ICTs have increased substantially in past few years and requested forum to review the need for additional transformation capacity at Allahabad and Mainpuri substations. It was decided in the meeting that CTU may be asked to do the system studies based on present and future forecasted load.

In the above OCC meeting, NRLDC intimated forum that N-1 issue at 400/220kV Allahabad (PG) Sub-station was already highlighted in the operational feedback report of GRID-INDIA (Quarter 1 & 2) of 2022-23. Further, NRLDC highlighted that at present, there is no major N-1 issue at 400/220kV Mainpuri (PG) Substation.

CTU intimated OCC forum that they have asked POWERGRID to confirm whether space is available for 4th ICT each at Allahabad (PG) & Mainpuri (PG) Substation. Based on the load profile communicated by NRLDC and space availability for 4th ICT, CTU will further deliberate it in their consultation meeting for evolving transmission schemes' for augmentation of 400/220kV transformation capacity at Allahabad and Mainpuri sub-station.OCC forum asked POWERGRID to propose a SPS scheme at Allahabad Substation in consultation with UP till implementation of capacity augmentation. Details of Augmentation of ICTs at Allahabad and Mainpuri substation is as under:

A. Augmentation of 400/220 kV, 1x500 MVA (4th) ICT at 400/220 kV Allahabad (PG) substation

From the loading pattern of Allahabad ICTs (3x315 MVA), it is observed that it is above N-1 limit for sufficient duration of time in most of time in year (except Oct-Dec). Loading of ICTs have reached maximum of about 860 MW in summer season (July'22, April'23 & May'23). From studies, it emerged that on 3 no. of 315 MVA ICTs, loading of ICTs was more than 'N-1' compliance limit (>750 MW) in evening peak demand scenario of summer season. The loading pattern of Allahabad ICTs for past one year is as below (Fig 1):

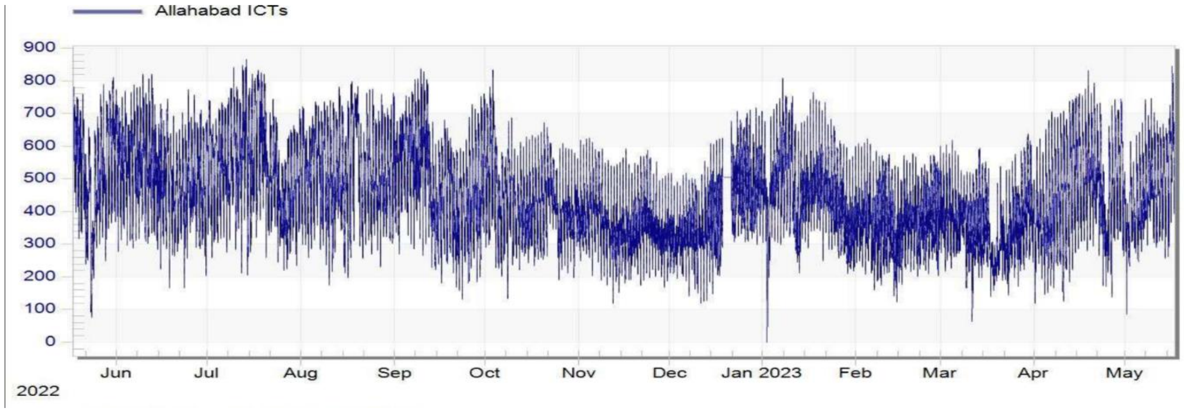

Fig 1: Loading of Allahabad ICTs of Past one year (Source: Grid-India)

Further, POWERGRID vide mail dated 22.06.23 confirmed space availability for Augmentation of 1x500 MVA (4th), 400/220kV ICT at Allahabad Substation. However 220kV side ICT bay will be AIS Type with ICT side termination through 220kV Cable/GIB

It was deliberated that in the studies, Short Circuit level at 400 kV Allahabad S/s is 47kA in present scenario which will further increase to 52kA in next 5 years against the designed capacity of 40kA. With augmentation of one 500MVA ICT, short circuit level will further increase by 1 kA. In view of this deliberation may be required to augment the ICT at Allahabad substation.

GRID-INDIA agreed for above proposal as the issue was already highlighted in operational feedback report . GRID-INDIA asked that whether any nearby lines i.e. 400kV Singrauli -Anpara is considered open or in service during fault calculation in studies. CTU informed that in planning studies all interconnecting lines were considered for short circuit calculation as line opening is a temporary solution to reduce the fault level. Grid-India informed that short circuit level of most of old substations in above complex (Singrauli/Rihand/Anpara) is high, however ICT augmentation is required with UP confirmation on above proposal as well as any intra state plan to relieve the loading of Allahabad ICTs. CTU stated that detailed studies will be carried out separately to address high short circuit levels in the complex.

UP informed that recently 400/220kV Jaunpur (1x315MVA +2x315MVA ICTs UC) substation is commissioned which relieved Allahabad ICT loading marginally, however overloading is already there at Allahabad substation. In addition to that UPPTCL also do not have any future plan in nearby area to relieve the ICT loading at Allahabad.

UP stated they have also carried out studies in Jun'24 timeframe in which 400/220kV Allahabad ICTs were observed to be overloaded and therefore ICT augmentation is required at 400/220kV Allahabad substation to meet the demand from 400/220kV Allahabad S/s reliably.

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In view of above deliberations, following ICT augmentation scheme was agreed in ISTS:

- Augmentation of 400/220 kV, 500 MVA (4th) ICT at Allahabad S/s along with associated transformer bays (AIS)
- 220kV ICT side termination through 220kV Cable/GIB

B. <u>Augmentation of 400/220 kV,1x 500 MVA (4th) ICT at 400/220 kV Mainpuri (PG) substation</u>

From the loading pattern of Mainpuri ICTs (2x315+1x500 MVA), it is observed that the loading is not breaching N-1 limit for peak loading of ICTs (680 MW). Further from rolling plan studies (2027-28) also, ICT is not breaching N-1 limits . In 207th OCC meeting, NRLDC also highlighted that at present, there is no major N-1 issue at 400/220kV Mainpuri (PG) Substation.

The loading pattern of Mainpuri ICTs for past one year is as below (Fig 2) :

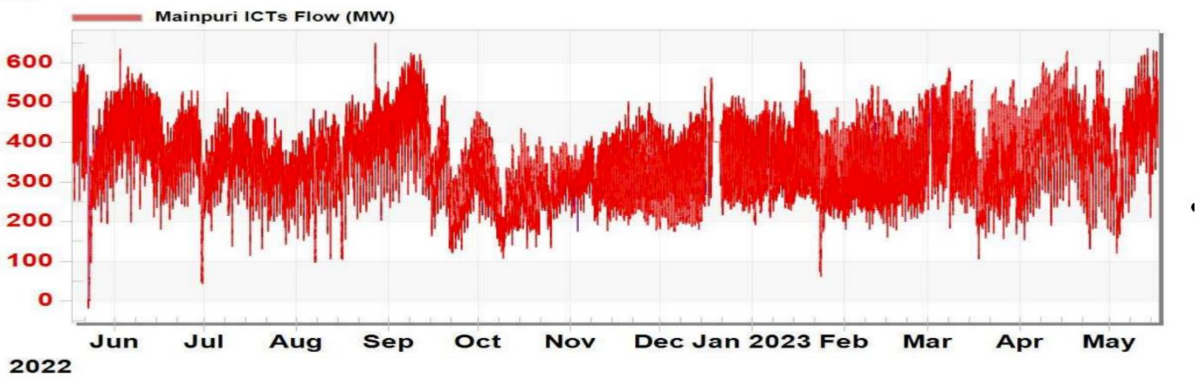

Fig 2: Loading of Mainpuri ICTs of Past one year (Source: Grid-India)

POWERGRID vide mail dated 22.06.23 confirmed space availability for Augmentation of 1x500 MVA, 400/220kV 4th ICT at Mainpuri Substation. However, it was informed that 125MVAr Bus Reactor needs to be shifted at future bay, accordingly bay equipment of above bay needs to be considered. 4th ICT will be placed at existing location of 125 MVAr Bus Reactor and 220kV side of ICT will be extended through 220kV GIB. 220kV side ICT bay will be Outdoor GIS Type with ICT side termination through 220kV GIB.

POWERGRID informed that loading of Mainpuri ICT increased substantially in past few years and therefore ICT augmentation is required. UP also requested them in this regard.

Grid-India informed that at present there is no requirement of 400/220kV ICT augmentation at Mainpuri (PG) substation, however UP views may be taken for ICT augmentation requirement to meet the future load in Mainpuri area as well as any future intra schemes planned/under implementation to divert the load of Mainpuri(PG) substation. UP stated that in future one no. 220kV line is under planning for drawl from Mainpuri substation. UP also stated that 400/220kV Farrukhabad is approved recently which will also relieve the loading of ICTs at Mainpuri(PG) substation and therefore at present there is no requirement for 400/220kV ICT augmentation at Mainpuri (PG) substation.

In view of above deliberations, operational data and studies indicated that at present there is no requirement of 400/220kV ICT augmentation at Mainpuri (PG) substation. However based on Grid-India operational feedback/real time data as well as future load growth in above area, proposal of ICT augmentation will be taken up at later stage.

2. Reconductoring of 220 kV Hisar (PG) - Hisar (IA) D/c line

It was deliberated that in 3rd NRPC-TP meeting held on 19.02.21, HVPNL's proposal of creation of 220 kV Chickenwas substation by LILO of both circuits of 220 kV Hisar (PG) to Fatehabad (HVPNL) D/c line was deliberated. In above agenda, Grid- India expressed concern on the high loading of 220 kV Hisar (PG) - Hisar (IA) D/c line and asked HVPNL to explore measures to relieve loadings of these lines. HVPNL agreed to the same.

Further, in the 54th NRPC meeting held on 31.05.22 it was desired that HVPN will take up the matter with CTU/CEA for reconductoring of 220 kV Hisar (PG) - Hisar (IA) D/c line and Connector to be replaced at Hisar (IA) end.. Further, HVPN vide letter dated 21.09.22 has suggested that 220 kV Hisar (PG) Hisar (IA) D/c line may be augmented with high performance conductor and matter of augmentation may be taken up with POWERGRID being the custodian of above line.

It was informed that Grid-India vide mail 21.06.22 shared the loading data of past one year (Jun'21-Jun'22)

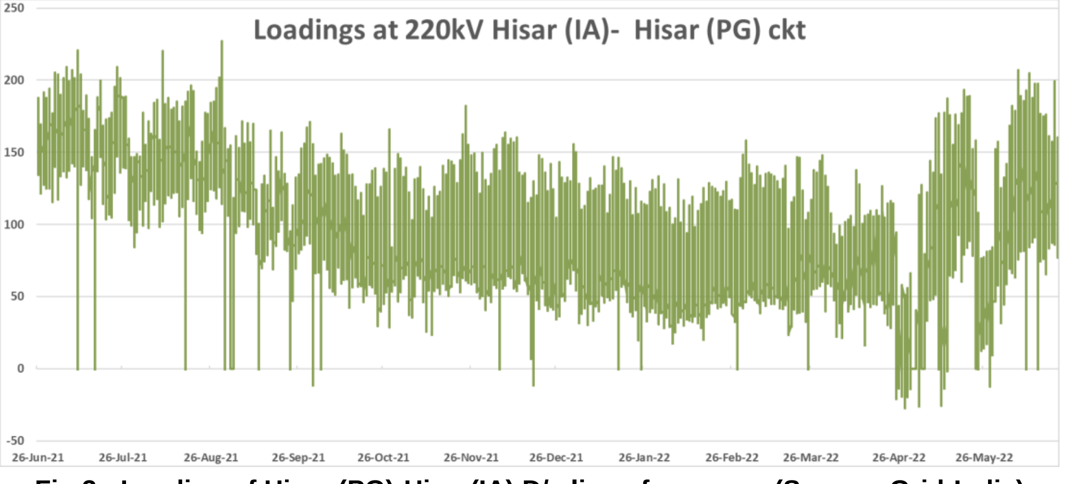

Fig 3 : Loading of Hisar (PG)-Hisar(IA) D/c line of one year (Source: Grid-India)

CTU vide mail 03.02.23 to CEA & Grid-India informed that as per the loading pattern of 220kV Hisar(PG)-Hisar IA-Hisar (BBMB) D/c line also , it is seen that loading of Hisar (PG) - Hisar (IA) is high and may not fulfill N-1 Criteria. Further, in the 62th NRPC meeting held on 31.01.23, NRLDC agenda for high loading of Hisar (PG) - Hisar (IA) was also deliberated and it was decided to resolve the above issue. In view of that, CTU requested in above mail to convene a joint meeting with HVPN/POWERGRID/Grid-India to discuss and finalize on possible alternatives (Reconductoring/ load diversion/220kV new interconnection) to resolve above high loading issue.

Subsequently a joint meeting was held on 20.03.2023 to discuss transmission related issues in Haryana incl. high loading issue of 220 kV Hisar (PG) - Hisar (IA) D/c line. In the meeting, Reconductoring of 220 kV Hlisar (PG) - Hisar (IA) D/c line with HTLS conductor was agreed. In the meeting, it was decided that POWERGRID will intimate the type of HTLS conductor which can be implemented on existing towers of 220 kV Hisar (PG) - Hisar (IA) D/c line within two weeks. It was also decided that HVPNL will carry out the augmentation of line bay equipment at 220kV Hisar (IA) substation in the matching time frame of reconductoring of the 220 kV Hisar (PG) - Hisar (IA) D/c line.

Subsequently, POWERGRID vide mail 25.05.23 informed to CTU that they need the Conductor Ampacity requirement for the subject line for deciding the technical parameters for HTLS conductor which is suitable for existing towers. CTU in reply vide mail 02.06.23 informed that from the loading data of one year (Jun’21-Jun’22), it is observed that maximum loading on 220 kV Hisar (PG) – Hisar(IA) D/c Line is about 225MW/ckt (under n-1 contingency loading may increase to 365MW). Considering above historical loading data, minimum conductor ampacity required to be about 1000 Ampere/ckt (about 381MW/ckt) for reconductoring of above 220kV line.

POWERGRID vide mail 09.06.23 mentioned that reconductoring of existing transmission line having ACSR Zebra conductor is feasible with HTLS conductor (equivalent to Zebra) to meet 1000 ampere current requirement with existing tower design.

CTU enquired HVPNL to update the status of measures taken to relieve the high loading on 220kV Hisar (IA) - Hisar (BBMB) D/c line (HVPNL).

Grid-India informed that reconductoring of above line with HTLS conductor of minimum 1000 ampere current requirement is adequate, however HVPN views may be taken for any future drawl requirement. HVPNL informed that at present, no future load is envisaged for drawl of power from Hisar (PG) substation.

POWERGRID informed that upto 1000 Ampere requirement, ACSS/GAP conductor were used in past for reconductoring of 220kV lines and same can be used for upto 1100 Ampere loading requirement. For requirement for more than 1100 Amperes, composite core type conductor is suitable due to sag limitation, however same is costlier (about 3 times) than ACSS/GAP conductor. CTU enquired about thermal capacity of ACSS/GAP type HTLS conductor available for more than 1000 Ampere (say 1050 A) with ACSS configuration.

POWERGRID stated that in the past they had the latest design of 1092 Ampacity conductor so 1050 Ampere HTLS conductor can be designed for reconductoring of above line. CTU stated that with 1050 Ampere (about 400MW/ckt) HTLS conductor shall further provide additional margin on 220 kV Hisar (PG) - Hisar (IA) D/c line to meet the future demand. Grid-India agreed for the same.

CTU stated that HVPNL to carry out the necessary bay equipment upgradation works at 220kV Hisar (IA) end in the matching time frame of reconductoring of the 220 kV Hisar (PG) - Hisar (IA) D/c line. HVPNL stated that at present Bay equipment are designed with1250 ampere rating at Hisar (IA) end, but conductor need be changed for tie bay. Additionally, the CT ratio needs to be checked and changed accordingly from 800 ampere to 1200 ampere. HVPN stated that they will do detailed analysis in consultation with design team to upgrade the bays equipment's at Hisar (IA) end in coordination with implementing agency for reconductoring works.

Further in the earlier joint meeting, it was deliberated that HVPNL will examine the issue of high loading (of similar order as 220kV Hisar(PG) – Hisar(IA) D/c line) on Hisar (IA) - Hisar (BBMB) 220 kV D/ c line and plan adequate measures to relieve loading on the line.

In above Joint meeting HVPNL informed that Zebra conductor may be used for above line, however after the detailed analysis it emerged that 220 kV Hisar (IA) - Hisar (BBMB) D/ c line is already augmented in 2009 with INVAR conductor. Loading on above line is also significantly reduced with implementation of 400kV Bhiwani (PG) - Bhiwani line. CTU enquired about Ampacity of augmented conductor of Hisar (IA) - Hisar (BBMB) 220 kV D/ c line. HVPNL informed that INVAR conductor (TASCR) in above line is designed for150 degree with about 960 Ampere capacity.

Grid-India informed that in past during high demand season (Jun-Aug), in the event of shutdown of one ckt of Hisar (IA) - Hisar (BBMB) 220 kV D/c line, SLDC instructed to take other ckt out of service also due to very high loading. Grid-India stated that with INVAR conductor, other ckt can carry the load in the event of shutdown of one ckt. HVPNL concurred on the same.

CTU requested HVPNL to re-check bay equipment rating at Hisar (IA) & Hisar (BBMB) end for Hisar (IA) - Hisar (BBMB) 220 kV D/c line for optimal utilization of above line and take the necessary action in case of upgradation requirement & also send a confirmation to CEA & CTU in this regard

In view of above following ISTS transmission strengthening scheme was agreed in the meeting

- **Reconductoring of 220 kV Hisar (PG) - Hisar (IA) D/c line (Single Zebra) with HTLS conductor (with minimum 1050Ampere/ckt requirement) along with bay equipment upgradation at 220kV Hisar (PG) end**

For above reconductoring works, HVPNL to carry out the necessary bay upgradation works in Intra state in the matching time frame of above agreed ISTS scope

3. Modification in Transmission scheme for Rajasthan REZ Ph-IV (Part-2 :5.5GW) (Jaisalmer/Barmer Complex)

Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :5.5GW) (Jaisalmer/Barmer Complex) was agreed in the 18th CMETS-NR meeting held on 28/04/23. The scheme was further discussed & approved in in 14th NCT meeting held on 09.06.23. In the meeting, based on the decision in 14th NCT meeting held on 09.06.23, following modifications was incorporated in present scope of transmission scheme in “Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :7.5GW) (Jaisalmer/Barmer Complex)”

- Replacement of 330MVAr line reactor with 240MVAr at Rishabdeo end on 765 kV Rishabdeo-Mandsaur D/c line and deletion of spare 110 MVAr reactor unit at Rishabdeo S/s.

Same was noted in the meeting.

List of Participants of 20th Consultation meeting for Evolving Transmission Schemes in NR held on 30.06.2023

CEA

Shri Nitin Deswal	Deputy Director

SECI

Shri Kaustuv Roy	GEM
Shri R.K.Agarwal	Consultant

Grid India

Shri Gaurav Malviya	Manager

NRPC

Shri Pradeep Kumar	Executive Engineer
Shri Rajat Dixit	Assistant Director

CTU

Shri Jasbir Singh	CGM (CTU)
Shri Kamal Kumar Jain	GM (CTU)
Shri Kashish Bhambhani	GM (CTU)
Shri Sandeep Kumawat	DGM (CTU)
Smt. Ankita Singh	Ch. Manager (CTU)
Shri R Narendra Sathvik	Manager (CTU)
Shri Roushan Kumar	Engineer (CTU)
Shri Madhusudan Meena	Engineer (CTU)

PGCIL

Shri Gunjan Agrawal	GM

Shri Mani Shekhar	DGM

HVPNL

Shri Sanjay Arora	CE (PD&C)
Shri Sandeep Yadav	

UPPTCL

Shri Satyendra Kumar	SE

PSTCL

Shri Nitin Kumar	Sr. XEN/Planning-1

PTCUL

Shri Ashok Kumar	Executive Engineer

GNA/Connectivity Applicants

Shri Gourav Soni	Green Infra Clean Wind Technology Limited
Shri Dhir Singh	Green Infra Clean Wind Technology Limited
Shri Vishnu Khandelwal	Hindustan Zinc Limited
Shri Pavan Kumar Gupta	Juniper Green Cosmic Private Limited
Shri Tushar Garg	O2 Power Private Limited
Shri Bhargava Sharma	Project Nine Renewable Power Private Limited
Shri Arzaan Dordi	Serentica Renewables India Private Limited
Smt Poorva Pitke	Sprng Power Private Limited
Shri Avinash Mirajkar	Sprng Power Private Limited